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REVIEW

Enhancing satellite-to-ground communication using quantum key distribution

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Abstract

Classical Cryptography has been in use for a long time. It has been the only way of securing people's communication. However, there are some flaws observed during the execution of classical cryptography. One of them being the staunch belief that the number composed of multiplication of two large prime numbers cannot be factorised easily, which is under a threat thanks to the computational power of the quantum computers. The next flaw is the non-detection of the hacker, both of which can be eliminated by using quantum mechanical principles for encryption purposes, which is known as quantum cryptography. Quantum Key Distribution, which provides an information-theoretically safe solution to the key exchange problem, is the most well-known example of quantum cryptography. The benefit of quantum cryptography is that it makes it possible to perform a variety of cryptographic operations that have been demonstrated or are hypothesised to be impractical when using solely traditional (i.e., non-quantum) communication. Free-space quantum communication has been successfully demonstrated across 300 m by the Indian Space Research Organization (ISRO) in March 2021. With this, ISRO is trying to achieve the same using a satellite-based communication mechanism, which would revolutionise the mode of modern communication. It is justified that the key generation rate depends on factors like the aperture diameter of the sender and receiver, distance between them, the quantum bit error rate, and many more. The results vary with the parameters in the discussion as explained in the upcoming sections. The avenue of different types of losses that occur while transmitting at large distances, such as Atmospheric Loss, Pointing Loss and Geometric Loss, is explored.

KEYWORDS

cryptography, cryptography protocols, quantum communication, quantum cryptography, quantum information

1 | INTRODUCTION

The two main disciplines of traditional cryptography are public key cryptography, commonly referred to as asymmetric cryptography, and secret or symmetric key cryptography [1]. The oldest type of cryptography is secret key cryptography, in which two parties share a secret key both to encrypt and decrypt their communications. Despite the fact that some secret key methods, such one-time pads, completely secure against arbitrary computational capability, they have a significant practical drawback that two parties must first establish a secure connection before they can communicate [2]. Secret Public key cryptography-based key distribution systems like Diffie-Hellman are used to create a secret key via an unsecure channel [3, 4]. In contrast to secret key cryptography, a shared secret key does not need to be established prior to communication in public key cryptography [5]. Instead, each party has a private key, which remains secret, and a public key, which they may distribute freely. If one party, say Alice, wants to send a message to another party, Bob, she would encrypt her

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message with Bob's public key after which only Bob could decrypt the message using his private key [6]. While there is no need for key exchange, the security of public key cryptography algorithms is currently all based on the unproven assumption of the difficulty of certain problems, such as integer factorisation or the discrete logarithm problem [7].

This means that public key cryptography algorithms are potentially vulnerable to improvements in computational power or the discovery of efficient algorithms to solve their underlying problems. Indeed, algorithms have already been proposed to perform both integer factorisation and solve the discrete logarithm problem in polynomial time on a quantum computer [8]. While the advent of a feasible quantum computer would make current public key cryptosystems [9] obsolete and threaten key distribution protocols, such as Diffie-Hellman, some of the same principles that empower quantum computers also offer an unconditionally secure solution to the key distribution problem [10]. Moreover, quantum mechanics also provides the ability to detect the presence of an eavesdropper who is attempting to learn the key, which is a new feature in the field of cryptography [11]. Because the research community has been focussed primarily on using quantum mechanics to enable secure key distribution [12], quantum cryptography and QKD are generally synonymous in the literature [13].

In this paper, we propose a method to enhance the satellite-to-ground communication by distributing the keys using QKD. This would allow various space organisations to make their communication reliable while sending private data across the communication channel since our paper focusses on providing the reader with every parameterised outcome, such as distance, wavelength, transmitter and receiver aperture, Beam Quality Factor and Zenith Angle, whereas in [14] the key rate's dependency is shown on a single parameter distance. The objective of this paper is to make the reader understand the work done in the field of QKD right from the basics of qubits till the analysis of key parameters for different protocols and losses.

2 | INTRODUCTION TO QUBITS

The classical information, the most common type of information, is exchanged in the form of 0 and 1 s—which are known as bits [15]. These form the basic units of information, that is, whatsoever the information may be, it can be expressed in a string of 0 and 1s. But the more people gained exposure to all of this, they found great loopholes in this which made the data of the consumer less secure. This made us look into alternatives, which can make the information exchange in a secure manner [1]. That is where Quantum Communication comes into picture. Over here, the basic units of information are termed as Qubits (Quantum Bits) [2]. The qubits can exist in a state of both 0 and 1 at the same time, which is known as superposition. This helps us to increase the computational space exponentially. To explain this, let us take an example.

Suppose at a grand party with many people, a person meets them, and as a sign of respect, shakes hands with everyone he meets. Let us say he shakes hands with 10 people in 5 min, but someone busier than the person shakes hands with 1024 (2^{10}) people in 5 min. The first person is representing a classical computational space where in any memory location either 0 or 1 can exist, but the second person is representing a quantum computational space where in any memory location both 0 and 1 can exist as seen in Figure 1.

Now, this was the advantage in computational spaces. What does it have to do with information exchange? [5] For that, we will understand about the state of a qubit by referring to Figure 2 known as a Bloch Sphere.

This is used to represent any quantum state ψ, in terms of a linear combination of $|0\rangle$ and $|1\rangle$, where $|0\rangle$ and $|1\rangle$ are the quantum equivalent states of 0 and 1. The representation is known as the key notation commonly used for representing quantum states [6]. In this, $|\psi\rangle$ can be expressed as $\alpha|0\rangle + \beta|1\rangle$, α and β being the constants having the constraint $|\alpha|^2 + |\beta|^2 = 1$. This indeed tells us that the radius of the Bloch sphere is 1 unit. Now, when we measure a quantum state, it collapses to a state of $|0\rangle$ or $|1\rangle$, the probability of occurrence for this depends on α and β [7, 8]. Suppose, we need to find the probability of occurrence of $|0\rangle$ in the quantum state $|\psi\rangle$ denoted as $\alpha|0\rangle + \beta|1\rangle$. Therefore, the probability is defined

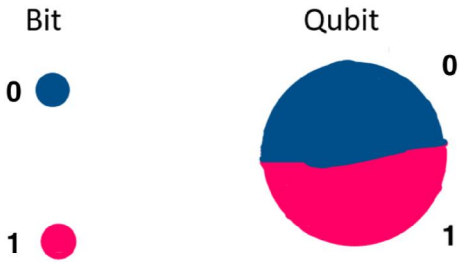


FIGURE 1 Classification of bit and qubit.

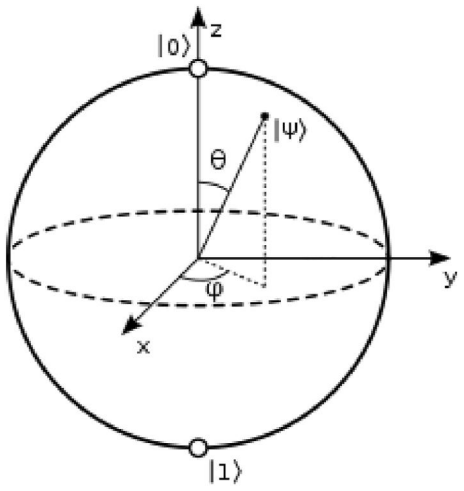


FIGURE 2 Bloch sphere.

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as $|\langle 0|\psi\rangle|^2 = |\alpha|^2$. Similarly, to find the probability of occurrence of $|1\rangle$, $|\langle 1|\psi\rangle|^2 = |\beta|^2$

So, what is the operation we are performing in the above steps? This is known as an inner product, similar to the dot product that we perform in vectors. As said before, as we measure a quantum state, it collapses to a state of $|0\rangle$ or $|1\rangle$. In addition, if you measure the state again, it would give you $|0\rangle$. That means, even if we measure a state, we cannot get the exact quantum state back. This forms the basis for a popular theorem known as no-cloning theorem. It states that it is impossible to create an independent and identical copy of an arbitrary unknown quantum state [9, 10]. This allows us to state that quantum communication is theoretically advantageous as compared to classical communication methods. [11] We will discuss the different QKD protocols and their applications over optical fibres and satellite-to-ground channels in the upcoming sections [12, 13]. Also, present cryptographic systems, such as RSA, use the concept of multiplication of two prime numbers to generate the public key. This has been under the threats of quantum computers since Shor's Algorithm has proved to factorise numbers much more efficiently. [14] Hence, we need secure quantum communication methods as well.

Secret key cryptography represents the most traditional form of cryptography in which two parties both encrypt and decrypt their messages using the same shared secret key as shown in Figure 3. While some secret key schemes, such as one-time pads, perfectly secure against an attacker with arbitrary computational power, they have the major practical disadvantage, that is, before two parties can communicate securely, they must somehow establish a secret key that can be taken care of using QKD [16–20].

3 | QKD

QKD is a secure communication method for exchanging encryption keys that are only known between shared parties [17]. The communication method uses properties found in quantum physics to exchange cryptographic keys in such a way that is provable and guarantees security [16].

One of the phenomena which Quantum Communication exploits is Entanglement. Entanglement is the property in which two photons, if they are entangled, can affect the state of one another even after placing them at an infinite distance. This also means that if you are aware of the state of one photon, they are linked in such a manner that you can definitely deduce the state of the other photon. Because of which the individuality of the photon is lost since now they have become sort of a single [18] entity.

3.1 | Theory of QKD protocols

3.1.1 | BB84

The idea is to encode every bit of the secret key into the polarisation state of a single photon. Because the polarisation state of a single photon cannot be measured without destroying this photon (which is referred to as collapsing of a state), this information will be 'fragile' and not available to the eavesdropper. This leads to the increase in the secure means of communication. Any eavesdropper (called Eve) will have to detect the photon, and then she will either reveal herself or will have to re-send this photon [5, 7, 9]. This would lead Eve to send a wrong polarisation state, which could be detected easily, thus increasing the security.

We define a binary 0 as a polarisation of 0° in the rectilinear bases or 45° in the diagonal bases. Similarly, a binary 1 can be 90° in the rectilinear bases or 135° in diagonal bases. Thus, a bit can be represented by polarising the photon in either one of two bases as shown in Figure 4.

The QKD protocol involves repeating this process enough times that an eavesdropper has a negligible chance of getting away with this interception. The steps that are involved in the BB84 protocol are explained and depicted in Figures 5–8:

(i) Alice chooses a string of random bits,
: 011,011,101,010
And a random choice of basis for each bit:
++X++X+XXX+

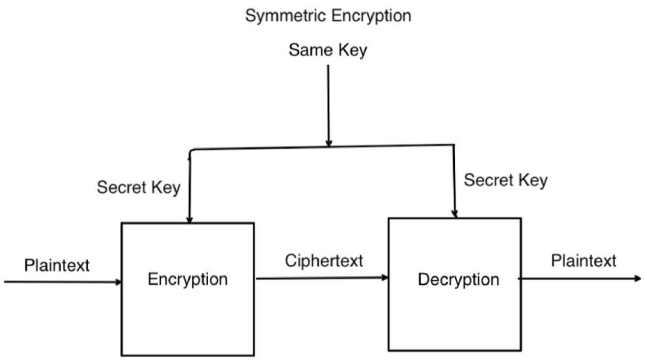


FIGURE 3 Symmetric encryption.

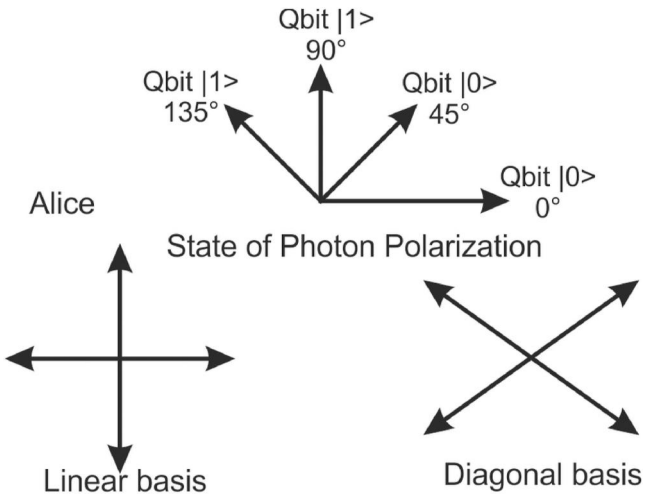


FIGURE 4 Polarisation states used in BB84.

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Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bits	0	1	1	0	1	1	1	0	1	0	1	0
Alice's random bases	+	+	X	+	+	+	X	+	X	X	X	+

FIGURE 5 Step 1—BB84.

Alice sends	→	↑	↖	→	↑	↑	↖	→	↖	↗	↖	→

FIGURE 6 Step 2—BB84.

Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Bob's random bases	X	+	X	X	+	X	+	+	X	X	X	+
Bob observes	↗	↑	↖	↖	↑	↗	↑	→	↖	↗	↖	→
Bob's bits	0	1	1	1	1	0	1	0	1	0	1	0

FIGURE 7 Step 3—BB84.

Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bases	+	+	X	+	+	+	X	+	X	X	X	+
Public channel	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕
Bob's random bases	X	+	X	X	+	X	+	+	X	X	X	+
Which agree?		✓	✓		✓			✓	✓	✓	✓	✓
Shared secret keys		1	1		1			0	1	0	1	0

FIGURE 8 Step 4—BB84.

Alice keeps these two pieces of information private to herself

(ii) Alice then encodes each bit onto a string of qubits using the basis she chose, this means each qubit is in one of the states $|0\rangle$, $|1\rangle$, $|+\rangle$ or $|-\rangle$, chosen at random. In this case, the string of qubits would look like this:

(iii) Bob also generates a set of basis and then measures each qubit using that basis, for example, he might use the bases: X+XX+X++XXX+. And Bob keeps the measurement results private.

(iv) Bob and Alice then publicly share which basis they used for each qubit. If Bob measured a qubit in the same basis Alice prepared it in, they use this to form part of their shared secret key, otherwise they discard the information for that bit.

(v) Finally, Bob and Alice share a random sample of their keys, and if the samples match, they can be sure (to a small margin of error) that their transmission is successful. If not, they redo this process. However, if Eve (the eavesdropper) has eavesdropped all the n bits of Alice and Bob, it reduces its probability of Eve being undetected with the relation $\frac{3}{4}^n$, which means if the sequence of bits is sufficiently long, chances of Eve getting to know the secret are less.

3.1.2 | Decoy protocols

The protocols that we have studied so far have been implemented on a single photon, so we need such laser sources, which could emit single photons. However, practically, we do not find such sources that could emit single photons but multiple photons, which lead to a very prominent attack known as Photon Number Splitting (PNS) Attack [12, 20]. Since, for a particular state, if more than one photon is generated, the hacker could deduce the state from the other photon generated for the state without being detected by Bob, which leads to privacy invasion [21].

Therefore, the concept of Decoy States was introduced. The meaning of Decoy is 'to deceive'. In this, we include decoy and vacuum states in addition to the signal state and transmit it with different mean photon numbers (denoted as μ), which is defined as the average photon number per transmitted pulse. This has been implemented on protocols like BB84 and B92 to enhance their security and include the issues brought down by practical cases as well.

Here is the sequence of steps done for BB84+Decoy protocol:

(i) Authenticate Partners—Authenticate partners with each other
(ii) Exchange Quantum Bits
(iii) Generate Sifted Key
(iv) Estimate Signal QBER—Accounts for signal state transmission errors
(v) Reconcile Errors—Uses bi-directional error correction algorithms
(vi) Detect Eavesdropper—Comparing yields of signal state, decoy state and expected yields in order to detect the eavesdropper
(vii) Estimate Entropy—Quantifying the amount of information leaked during previous QKD activities
(viii) Generate a Secret Key

Any Decoy protocol is configured according to these three parameters:

1) Signal State
   To transmit weak coherent pulses for generating shared secret key and facilitates higher key distribution by increase in mean photon number
2) Decoy State
   Responsible for detecting Photon Number Splitting Attacks in the channel
3) Vacuum State
   Responsible for measuring the dark count rate

4 | COMPONENTS IN A SATELLITE-TO-GROUND COMMUNICATION CHANNEL

In a Satellite-to-Ground Channel, the components across it are the Satellite and the Transmission Medium in between the sender and the receiver. This section would discuss this aspect in-depth.

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4.1 | Satellite

This houses the optical instruments as per the satellite's requirement in the experiment. This means that the satellite can act as a sender, a receiver or a relay device as well [6, 8]. The concept of using satellites as a relay device generally comes during entanglement-based QKD where we require an entangled photon source in addition to the two parties (Alice and Bob).

The position of the satellite with respect to the ground station makes a lot of differences as well [11, 14]. We consider using two of the satellite's configurations depicted in Figure 9:

1) LEO (Low Earth Orbit) Satellites
   These satellites usually move with very high speeds on low altitudes (500–1200 km), which reduce our total channel losses drastically. However, due to the high speeds, it becomes really difficult for the satellite and ground station to be properly aligned with each other, which induces shorter windows for conducting the experiment.
2) GEO (Geostationary Earth Orbit) Satellites
   These satellites move with the speed of the earth and thus they remain stationary with respect to the ground. But they are kept at an altitude of 36,000 km, which in turn bring in great amount of total channel losses.

The configuration of the satellite can be discussed with respect of the execution of the protocols as well [4, 22]. We can explain it under three conditions:

(i) Satellite as the sender and the ground station as the receiver. As shown in Figure 10, the satellite and the ground station perform the key distribution process and distribute the key K_a.
(ii) Satellite as the receiver and the ground station as the sender. As shown in Figure 11, the satellite and the ground station perform the key distribution process and distribute the key K_b.
(iii) Satellite as the relay device between two ground stations (Trusted Node) in Figures 12 and 13.
   (a) This means that the satellite first performs a key distribution between itself and the first ground station (Station A) and shares a key K_a. Then, the same

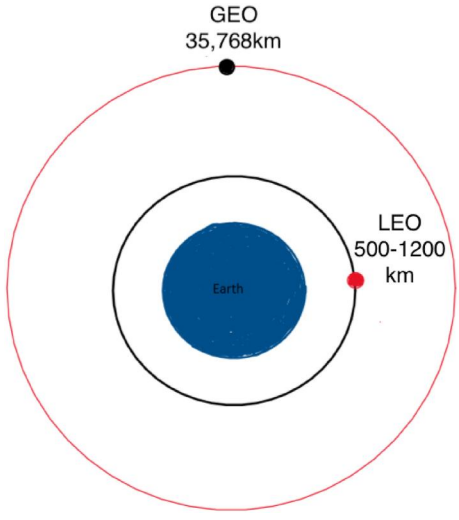


FIGURE 9 Low Earth Orbit (LEO) and Geostationary Earth Orbit (GEO) orbits.

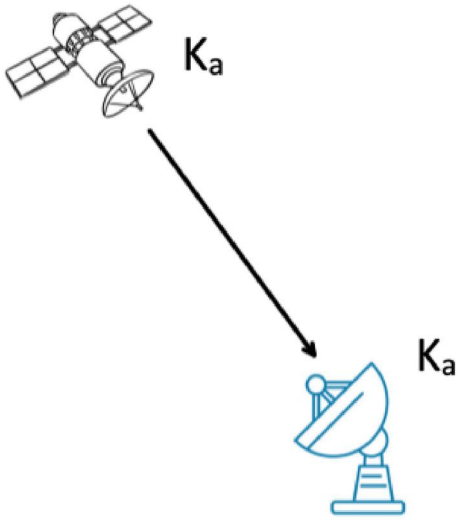


FIGURE 10 Satellite-to-Ground: 1st Configuration.

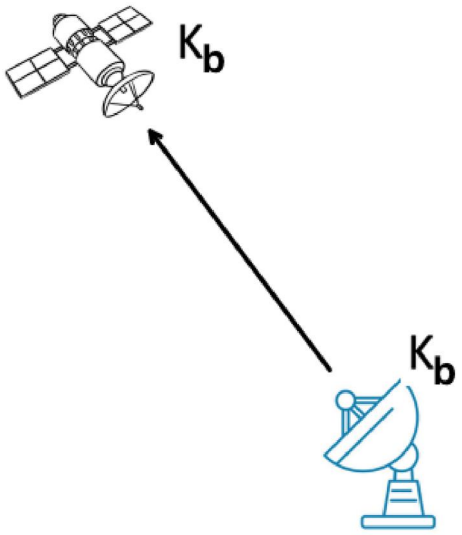


FIGURE 11 Satellite-to-Ground: 2nd Configuration.



FIGURE 12 Satellite-to-Ground: 3rd Configuration.

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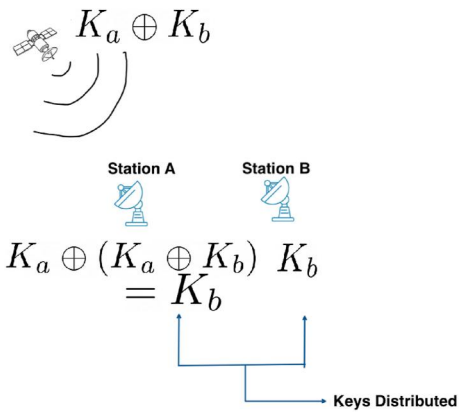


FIGURE 13 Satellite-to-Ground: 3rd Configuration continued.

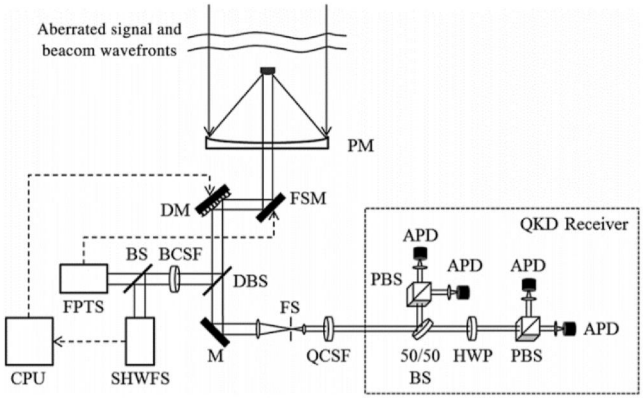


FIGURE 14 Adaptive optics setup.

satellite does perform a key distribution between itself and the second ground station (Station B) and shares a key K_b.

(b) It stores both keys and then performs a XOR operation and stores this result ($K_a \oplus K_b$). Then, it shares this to either one of the stations and lets them perform a XOR operation with the key they have. By this, we then ensure that both of the stations receive the same key. The reason we call this a trusted node-based system is because the satellite carries the information for both the keys, for which we trust the satellite to keep this information secret to any other party.

5 | TRANSMISSION OF SIGNAL

This is where most of the uncertainties lie. During the conduction of the experiment, when the laser passes through the atmosphere and vacuum, it encounters different types of losses as follows:

1) Atmospheric Loss
These losses occur due to the change in refraction of the air while the signal traverses a path through air, also the signal gets scattered due to the interference of particles.
One of the most popular models that we include to understand the strength of atmospheric turbulence is the Hufnagel-Valley model in Equation (1)

$$C_n^2(h) = 0.00594(v/27)^2\left(10^{-5}h\right)^{10}\exp(-h/1000) + 2.7 \times 10^{-16}\exp(-h/1500) + A\exp(-h/100) \quad (1)$$

where,

(i) h is the distance in metres
(ii) v is the pseudowind (m/s)
(iii) A is the nominal value of $C_n^2(0)$ at the ground in $m^{-2/3}$

Hufnagel-Valley Model with $v = 21$ m/s and $A = 1.7 \times 10^{-14}\text{m}^{-2/3}$ corresponds to the values for the Fried Coherence Parameter as 5 cm.

Fried Coherence Parameter is the measure of the quality of the optical transmission through the atmosphere due to random inhomogenities in the refractive index.

2) Pointing Loss
In an ideal scenario, the misalignment losses would be 0 dB, which implies that the ground station achieves maximum gain. This happens when both the transmitter and receiver antenna are 100% aligned. However, as we have seen for Low Earth Orbit Satellites, the satellite speed is huge because of which the alignment between the satellite and the ground station is not seen at all times. One of the architectures that has been quite successful in lowering the pointing losses is the use of Adaptive Optics [22].

The components shown in Figure 14 are:

(a) Fast Steering Mirror (FSM)
Tracks the tilt component of the wavefront error
(b) Deformable Mirror (DM)
Compensates Higher Order Wavefront Errors
(c) Dichroic Beam Splitter (DBS)
Diverts the beacon wavefront to the wavefront sensing system
(d) Beacon Channel Spectral Filter (BCSF)
Transmits the beacon wavelength while blocking other spectral components
(e) Focal Plane Tracking Sensor (FPTS)
Measures the tilt component of the wavefront error and generates control signals for the FSM
(f) Beam Splitter (BS)
Transmits a portion of the beacon light to FPTS (Focal Plane Tracking Sensor)
(g) Shack-Hartmann Wavefront Sensor (SHWFS)
Determines the Higher-Order aberrations and generates control signals for the DM
(h) Quantum Channel Spectral Filter (QCSF)
Transmits the Quantum Channel wavelength to the QKD receiver while blocking other spectral components

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Also, it is preferred to have satellite-to-ground communication (Downlink) rather than ground-to-satellite (Uplink) since the stream of photons in the satellite-to-ground channel has an advantage of travelling in the vacuum (space) in its initial stages, which leads to a low amount of losses, whereas in the ground-to-satellite channel, it has to travel in the earth's atmosphere first, which leads to huge losses.

6 | PRACTICAL IMPLEMENTATION AND RESULTS

In this section, we will discuss the implementation of BB84, Decoy BB84 and Geometric Loss with respect to parameters like distance, diameters of the receiver and transmitter mirror, zenith angle, and wavelength.

6.1 | BB84

When a QKD protocol is being implemented, there are some parameters that should be understood, which are not included in the theoretical discussions of these protocols [6, 7, 11]. With this, the parameters associated with BB84 are as follows:

- e_n—QBER for n-photon system
  QBER or Quantum Bit Error Rate is the percentage of error measured during the experiment. To understand the concept of QBER, here is an example:—Let us say 10 qubit states have been transmitted, out of which due to the irregularities faced by the transmitted qubits (could be due to detectors inefficiency, hampering in the transmission channel), the state of two qubits is disrupted. Therefore, the error is observed in 2 qubits out of the 10 qubits or 20%. This is known as the Quantum Bit Error Rate or commonly abbreviated as QBER.
  For an ideal single-photon source, the QBER is given by Equation (2):

$$e = \frac{(0.5 * p_{dark} + e_{\text{det}} * p_{signal})}{(p_{signal} + p_{dark})} \tag{2}$$

  where e_{det} is the error caused due to the detector, p_{signal} for this is equated as the transmission efficiency of the channel and p_{dark} is the inefficiency parameter of the detector to detect a signal even if there is not a signal incident on it. This mainly arises due to influences of the surroundings.
  However, for a non-ideal/practical source, the QBER for an n-photon system is defined by Equation (3)

$$e_n = \left(\eta_n \frac{e_{\text{det}}}{2} + (1-\eta_n) p_{dark} \frac{1}{4}\right) / Y_n \tag{3}$$

  where η_n and Y_n are the Transmission efficiency for an n-photon system and the yield for an n-photon system, respectively.

- Y_n—Yield for n-photon system
  This represents the conditional probability that Bob would detect the signal transmitted by Alice. The expression for the yield is given by Equation (4):

$$Y_n = [\eta_n + (1-\eta_n) p_{dark}]/2 \tag{4}$$

- Q_n—Gain for n-photon system
  This is the average yield for the n-photon system given in Equation (5):

$$Q_n = Y_n e^{-\mu} \mu^n / n! \tag{5}$$

  Since ideal single photon sources are not available, we focus our research on weak coherent pulse sources where we tend to generate a stream of single photons according to a probabilistic Poisson distribution [12]. We make this possible using the concept of mean photon number (commonly termed as μ), which is defined as the average photon number per transmitted pulse.
- η—Total Efficiency
  This is calculated keeping the channel (termed as u) and detector loss into consideration given in Equation (6):

$$\eta = 10^{(-u/10)} \times 0.5 \tag{6}$$

- η_n—Transmission efficiency of the n-photon system
  It includes the dependency of efficiency with photon numbers as well. This is given in Equation (7):

$$\eta_n = 1 - (1-\eta)^n \tag{7}$$

- Q_μ—Overall gain of the system
  It is the weighted average of the yields calculated before in Equation (5), which is given in Equation (8):

$$Q_\mu = Y_0 e^{-\mu} + Y_1 e^{-\mu} \mu + Y_2 e^{-\mu} \frac{\mu^2}{2} + \ldots Y_n e^{-\mu} \frac{\mu^n}{n!} \tag{8}$$

- E_μ—Overall QBER of the system
  It is defined as the weighted average of the errors in the system given in Equation (9):

$$E_\mu = \frac{1}{Q_\mu} \sum_{n=0}^{\infty} Y_n e^{-\mu} \left(\frac{\mu^n}{n!}\right) e_n \tag{9}$$

- R—Key Rate of this system
  Key Generation Rate is the parameter that defines how well the key is going to be generated amongst the two parties. It depends on several parameters, such as QBER, Gain, Transmission Efficiency, Error Correction Efficiency and so on.
  For an ideal single-photon source, the key generation rate is given in Equation (10):

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$$R = S/2 \cdot [1 - f_e H_2(e) - H_2(e)] \quad (10)$$

where

- f_e is the error correction efficiency, which is assumed to be 1.22
- S is the probability that a non-empty pulse is detected by Bob, which is also equated as the transmission efficiency of the channel and
- $H_2(x)$ is the binary entropy calculated by Equation (11)

$$H_2(x) = -x * \log_2(x) - (1 - x) * \log_2(1 - x) \quad (11)$$

Whereas, for a non-ideal/practical source, the key generation rate is given by Equation (12):

$$R = -Q_\mu f_e H_2(E_\mu) + Q_1[1 - H_2(e_1)] \quad (12)$$

6.2 | Secure Key Rate (SKR) versus channel loss

This simulation relates the key rate with the channel loss for a non-ideal system. To make it possible, we have made use of Equation (12) and the relation of channel loss on total efficiency depicted in Equation (6) and the set of parameters used for this is shown in Table 1.

As depicted in Figure 15, we see that for values of $\mu = 0.2$ and 0.4, at a loss of 30 dB, the values of key rate are 1.42 kbit/s and 0.6 kbit/s for 100 MHz.

TABLE 1 Table of parameters.

Parameter	Values
f_e	1.22
p_{dark}	10^{-6}
e_{det}	0.01

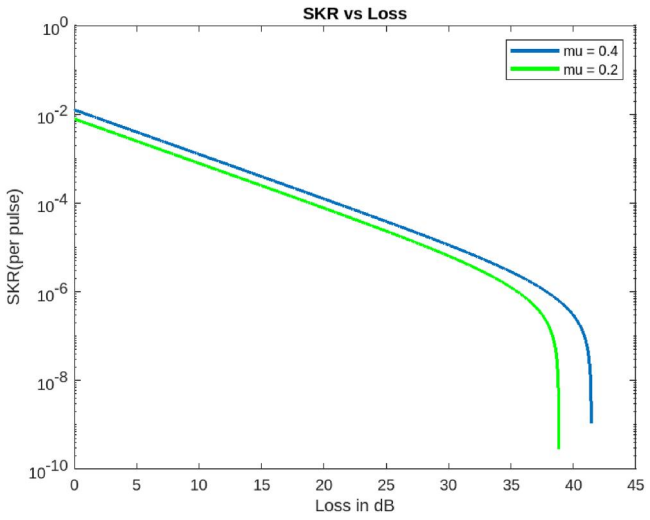


FIGURE 15 Secure Key Rate versus channel loss.

6.3 | QBER versus loss

[8] To generate this simulation, Equation (2) was used but assuming p_{signal} to be $1 - \exp(-\eta^*\mu)$, which states the non-ideal case since p_{signal} depends on μ. The parameters that were used to generate the simulation are given in Table 2. As shown in Figure 16, the QBER saturates at 0.5 for both the mean photon numbers; however, for higher mean photon numbers, the channel loss is low for a given QBER.

6.4 | Key rate v/s zenith angle

To make this simulation possible, a loose assumption was made that the effective length between the satellite and the ground station is now L*sec θ, where θ is the angle between the satellite and the ground station and L is the minimum distance between the satellite and the ground station.

The equations that were used to generate this simulation are Equation (13), which would be our loss parameter, and Equation (6) is used subsequently to find a relation between the angle and the key rate using Equation (12) to do so. The list of parameters for generating this simulation is given in Table 3.

As seen in Figure 17, it can be deduced that the zenith angle has an immense effect on the key rate. The rate is maximum when the satellite is just above the ground station (at $\theta = 0$ rad), and the rate decreases with an increase in the zenith angle.

TABLE 2 Table of parameters.

Parameter	Values
p_{dark}	10^{-3}
e_{det}	0.05

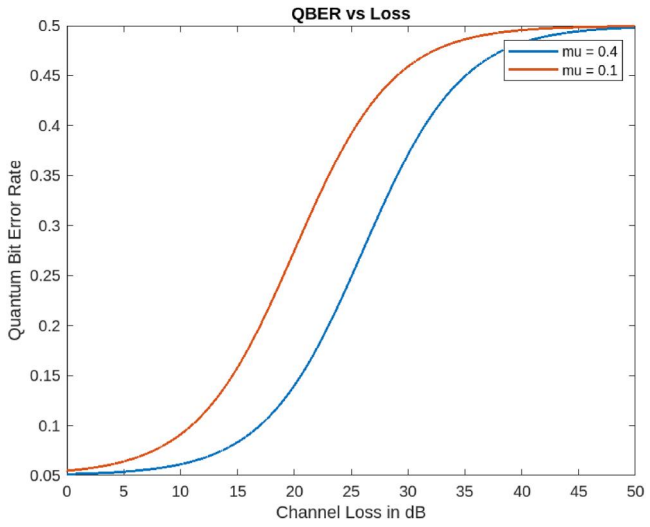


FIGURE 16 Quantum Bit Error Rate (QBER) versus loss.

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TABLE 3 Table of parameters.

Parameter	Values
p_{dark}	10^{-6}
e_{det}	0.01
D_r	1 m
D_t	0.3 m
m	1.318
λ	780 nm
L	500 km
μ	0.3
f_e	1.22

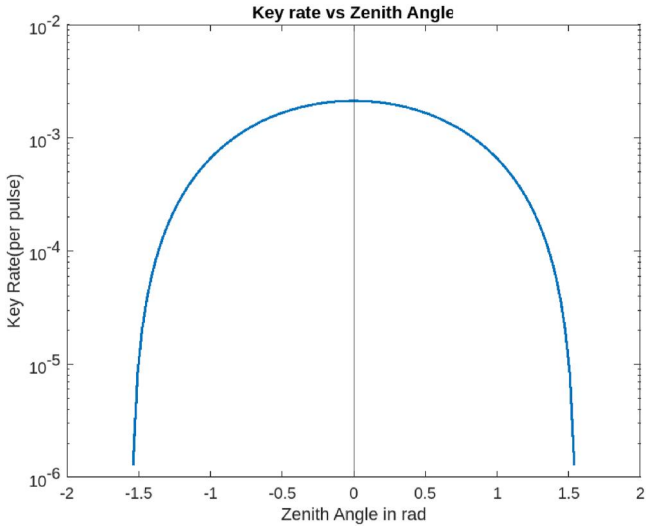


FIGURE 17 Key rate (per pulse) versus zenith angle.

6.5 | Geometric loss

Geometric loss, or diffraction loss, refers to the loss due to the beam spread effect when it propagates from the satellite to the optical ground station. [19] In order to explain this, consider that a torch is lit and kept around a 1 inch distance from a wall. We know that the light on the wall would be concentrated around a point and be bright. Now, when the distance between the torch and the wall increases, the light is scattered on the wall and the intensity of the light decreases.

[20] This beam spread accounts for huge losses since the detector's diameter on the receiver end would be small as compared to the large spread, which leads to loss in acquiring data. [23] A proper analysis is done in the later chapters, which gives us the parameters for acquiring low geometric loss [21]

The expression for geometric loss is depicted in Equation (13) [24]:

$$u = 20 * \log\left(\frac{D_r * D_t}{D_t^2 + 2.44 * m * x * \lambda}\right) \tag{13}$$

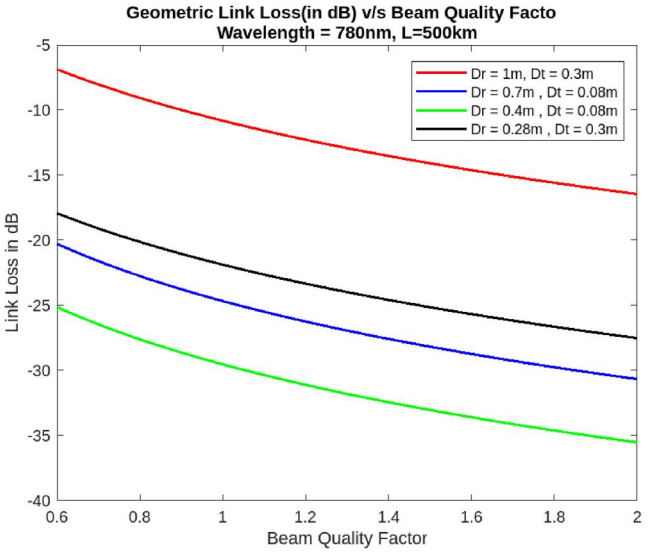


FIGURE 18 Geometric loss versus distance.

where

D_r represents the aperture diameter of the receiver.

D_t represents the aperture diameter of the transmitter.

m represents the Beam Quality Factor, which depicts how close the actual far-field divergence angle of the laser beam is to the theoretically perfect case.

x represents the distance between the transmitter and the receiver.

λ represents the wavelength of the laser being used for the conduction of the experiment.

6.6 | Geometric loss versus distance

Figure 18 shows that for $D_r = 1$ m and $D_t = 0.3$ m, minimum link loss is obtained, that is, in the range of -10 dB to -20 dB for distances from 500 to 1200 km. Therefore, these values can be used to design the mirrors of the transmitter and receiver to have minimum loss as per our distance requirements.

6.7 | Geometric loss versus Beam Quality Factor

Figure 19 shows us that for $D_r = 1$ m and $D_t = 0.3$ m, minimum link loss is obtained, that is, in the range of -5 dB to -17 dB for values of Beam Quality Factor from 0.6 to 2. Therefore, these values can be used to design the mirrors of the transmitter and the receiver to have minimum loss as per the dependence of Beam Quality Factor.

6.8 | Geometric loss versus wavelength

Figure 20 shows us that for $D_r = 1$ m and $D_t = 0.3$ m, minimum link loss is obtained, that is, in the range of -9 dB to

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−20 dB for wavelengths from 500 to 2000 nm. Therefore, these values can be used to design the mirrors of the transmitter and the receiver to have minimum loss as per our wavelength requirements.

6.9 | Receiver aperture v/s transmitter aperture for different losses at 500 km

Figure 21 shows a concise variation of the values in the transmitter and receiver apertures for different losses at a distance of 500 km. In order to obtain minimum link loss at a wavelength of 780 nm and Beam Quality Factor = 1.318, we must follow the values depicted in the purple section of the graph since the purple section denotes a loss of −15 dB.

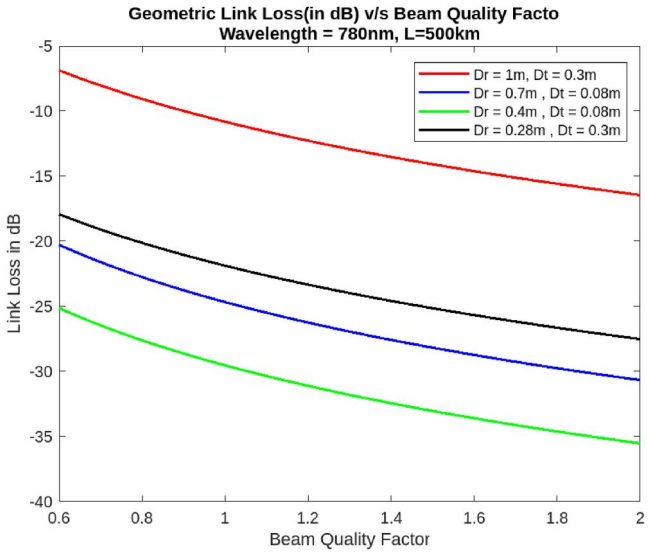


FIGURE 19 Geometric loss versus Beam Quality Factor.

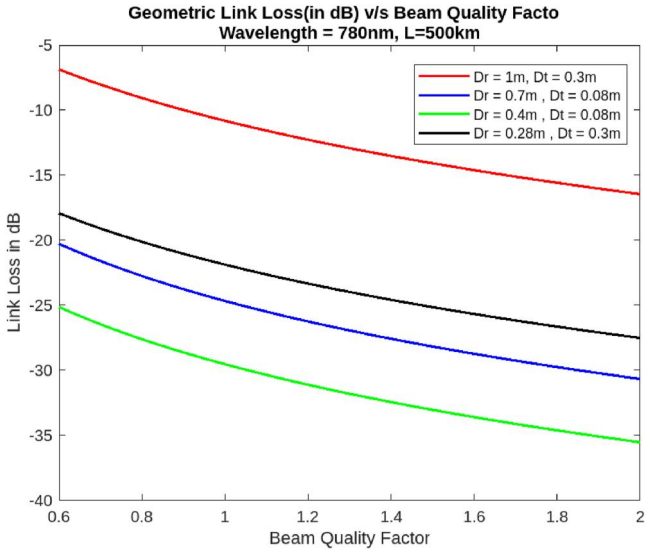


FIGURE 20 Geometric loss versus wavelength.

6.10 | Receiver aperture v/s transmitter aperture for different losses at 1000 km

Since the distance has increased to 1000 km, in Figure 22, the variation of transmitter and receiver aperture has changed. As observed in the graph, values for −15 dB loss in the graph are not completely visible at all which is why if we want to construct for 1000 km, we must opt for the values in the light blue section (−20 dB).

6.11 | Decoy BB84

- The gains (Q_μ, Q_{ν_1} and Q_{ν_2}) and QBERs [21] (E_μ, E_{ν_1} and E_{ν_2}) associated with the signal and decoy states are given in Equations (14)–(16):

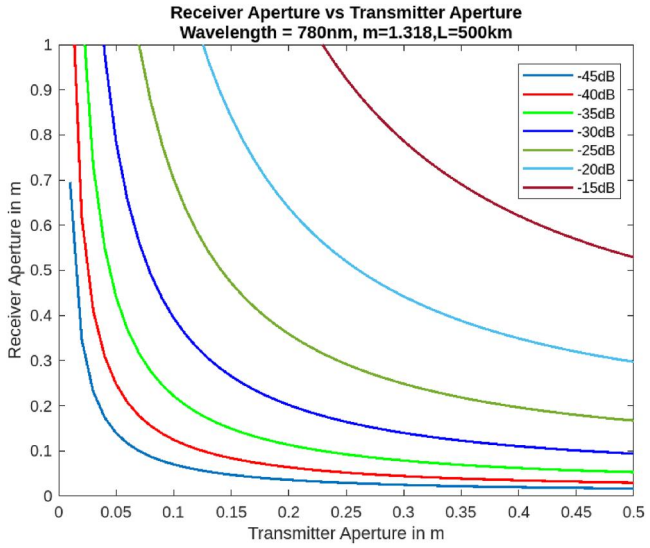


FIGURE 21 Receiver aperture v/s transmitter aperture for different losses at 500 km.

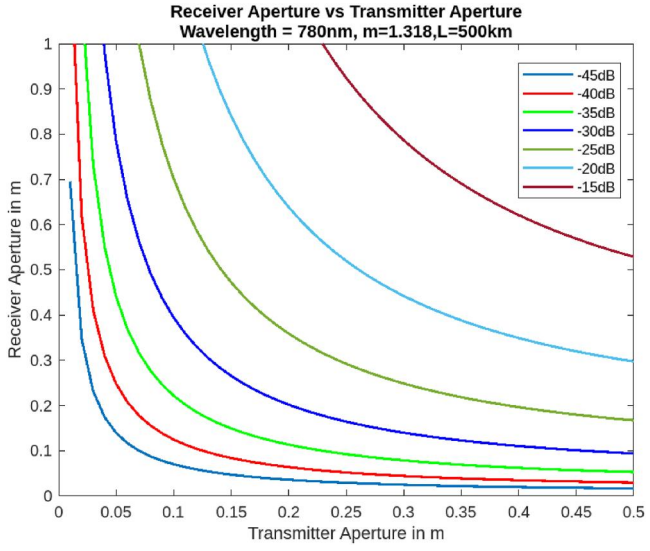


FIGURE 22 Receiver aperture v/s transmitter aperture for different losses at 1000 km.

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$$Q_\mu = \sum_{i=0}^{\infty} Y_i e^{-\mu} \mu^i / i!, E_\mu Q_\mu = \sum_{i=0}^{\infty} Y_i e_i e^{-\mu} \mu^i / i! \quad (14)$$

$$Q_{\nu_1} = \sum_{i=0}^{\infty} Y_i e^{-\nu_1} \nu_1^i / i!, E_{\nu_1} Q_{\nu_1} = \sum_{i=0}^{\infty} Y_i e_i e^{-\nu_1} \nu_1^i / i! \quad (15)$$

$$Q_{\nu_2} = \sum_{i=0}^{\infty} Y_i e^{-\nu_2} \nu_2^i / i!, E_{\nu_2} Q_{\nu_2} = \sum_{i=0}^{\infty} Y_i e_i e^{-\nu_2} \nu_2^i / i \quad (16)$$

where μ, ν_1 and ν_2 are the mean photon numbers for the signal, decoy and vacuum states, respectively.

- Lower bound of the background event yield Y_0 is depicted in Equations (17) and (18):

$$\nu_1 Q_{\nu_2} e^{\nu_2} - \nu_2 Q_{\nu_1} e^{\nu_1} \leq (\nu_1 - \nu_2) Y_0 \quad (17)$$

$$\Rightarrow Y_0 \geq Y_0^L = \frac{\nu_1 Q_{\nu_2} e^{\nu_2} - \nu_2 Q_{\nu_1} e^{\nu_1}}{\nu_1 - \nu_2} \quad (18)$$

- Lower bound of the single photon yield Y_1 is depicted in Equations (19)–(22):

$$Q_{\nu_1} e^{\nu_1} - Q_{\nu_2} e^{\nu_2} (\nu_1 - \nu_2) Y_1 + \sum_{i=2}^{\infty} Y_i \frac{(\nu_1^i - \nu_2^i)}{i!} \quad (19)$$

$$\leq (\nu_1 - \nu_2) Y_1 + \frac{\nu_1^2 - \nu_2^2}{\mu^2} \sum_{i=2}^{\infty} Y_i \frac{\mu^i}{i!} \quad (20)$$

$$\leq (\nu_1 - \nu_2) Y_1 + \frac{\nu_1^2 - \nu_2^2}{\mu^2} \left(Q_\mu e^\mu - Y_0^L - Y_1 \mu \right) \quad (21)$$

$$Y_1 \geq Y_1^L \frac{\mu}{\mu\nu_1 - \mu\nu_2 - \nu_1^2 + \nu_2^2} \times x \quad (22)$$

where $x = \left(Q_{\nu_1} e^{\nu_1} - Q_{\nu_2} e^{\nu_2} - \frac{\nu_1^2 - \nu_2^2}{\mu^2} \left(Q_\mu e^\mu - Y_0^L \right) \right)$

- QBER of the Decoy States as shown in Equation (23):

$$e_1 \leq e_1^U = \frac{E_{\nu_1} Q_{\nu_1} e^{\nu_1} - E_{\nu_2} Q_{\nu_2} e^{\nu_2}}{(\nu_1 - \nu_2) Y_1^L} \quad (23)$$

- Key Generation Rate depicted in Equation (24):

$$R \geq q \left\{ -Q_\mu f(E_\mu) H_2(E_\mu) + Q_1^L \left[1 - H_2(e_1^U) \right] \right\} \quad (24)$$

6.12 | Key rate versus distance

The simulation is generated by employing Equation (24) for finding the key rate in terms of distance, which would be plugged into the equation by employing Equation (13) as the loss parameter. The table of parameters used in generating this simulation is mentioned in Table 4.

As seen in Figure 23, the protocol is sustainable for distances up to 4500 km. The standard distance values along with the key rate are depicted in Table 5.

6.13 | Comparative analysis between BB84 and decoy BB84

To compare the sustainability of BB84 and Decoy BB84 with the channel loss, we will make use of Equations (10), (12) and

TABLE 4 Table of parameters.

Parameter	Values
p_{dark}	10^{-6}
e_{det}	0.05
D_r	1 m
D_t	0.3 m
m	1.318
λ	780 nm
L	500 km
μ	0.8
v_1	0.3
v_2	0
f_e	1.22

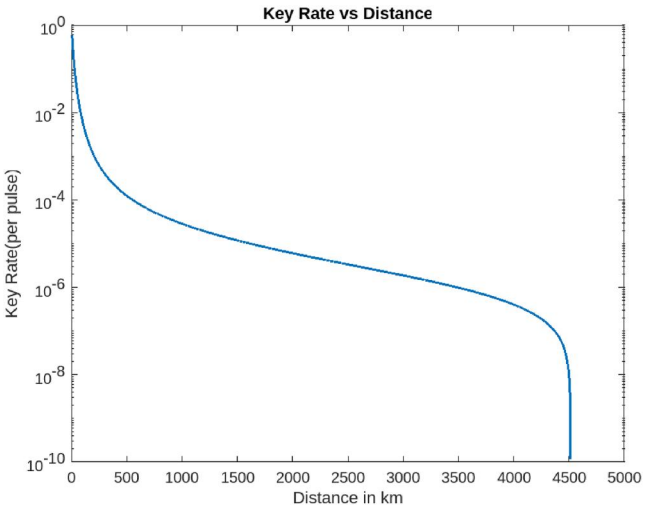


FIGURE 23 Key rate versus distance decoy BB84.

TABLE 5 Values of Key Rate with different lengths.

Length in km	Key rate@ 100 MHz
500	12.6 kbit/s
1000	2.88 kbit/s
2000	0.6 kbit/s

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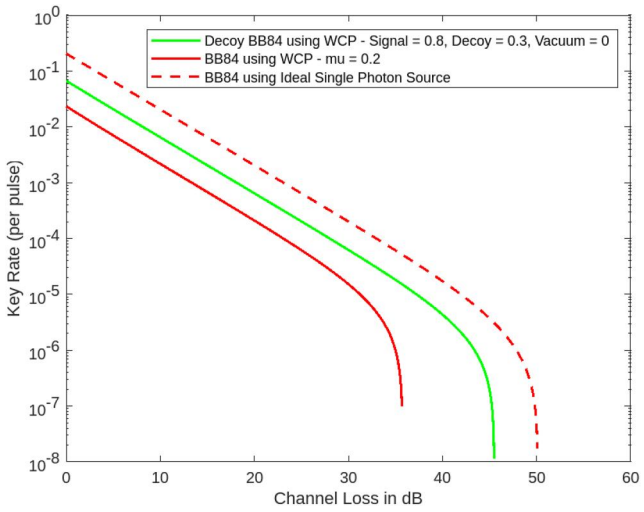


FIGURE 24 Comparison between BB84, decoy BB84 and an ideal BB84.

(24), which are the key rate equations for BB84 with an ideal photon source, BB84 with a non-ideal photon source and Decoy BB84.

As visible in Figure 24, the practical BB84 can sustain losses till 35 dB, whereas the practical Decoy BB84 sustain losses till 45 dB. This does show that for long distances where we need the protocol to be sustainable for large losses, Decoy BB84 is the clear winner.

Although the ideal BB84 is sustainable for losses till 50 dB, it isn't possible for us to manufacture photon sources which generate single-photons. Therefore, practically we still prefer Decoy BB84.

7 | CONCLUSION AND FUTURE SCOPE

The detailed simulations and analysis of QKD protocols for space-based communication scenarios have depicted a clear relation between parameters, like Secure Key Rate, Distance, Mean Photon Number, and Zenith Angle for QKD protocols. Depending on the use case considering these parameters, we can make sure that we get the maximum output from the whole execution of the protocol.

Not only for the protocols, but a clear understanding of the losses that is encountered by the transmitted photon stream is also presented. The dependence of loss on parameters, like distance, wavelength, and aperture diameters of transmitter and receiver, helps to understand the importance of each parameter and to select a perfect set of parameters that would lead to achieve low losses at the path of transmission. According to the results obtained against Geometric Loss, we have found that as the values for the diameter of the receiving and transmitting mirrors increase, the loss decreases. We have proved this statement to be true by seeing this for $D_r = 1$ m, $D_t = 0.3$ m as compared to other values kept under purview depicted in Figures 18–20.

Also, if the key rates for BB84 and Decoy BB84 are compared as opposed to the channel loss as shown in Figure 24, it is found that Decoy BB84 is a clear winner since it is sustainable for 45 dB losses as well. Therefore, Decoy BB84 can be used for applications over large distances.

The future holds a great promise to improve the techniques that are being used for implementing these protocols. QKD has been a great opening for the world to experience secure communication methods and it is an amazing avenue for everyone to explore. Scientists across the world have been constantly working on creating a robust mechanism that includes minimising losses so that this could reach a larger crowd instead of pertaining it to a certain group of eminent people. In the near future, it has the potential to get commercialised as well so that these enhanced security methods can reach many people.

AUTHOR CONTRIBUTIONS

All authors contributed equally.

CONFLICT OF INTEREST STATEMENT

The authors have declared no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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